

Week 1: Financial Asset Prices and Returns

Motivation

1. Much of finance is concerned with measuring and managing financial risk.
2. **Risk**: uncertainty in future returns from an investment.
3. Fin. Metrics is the study of random phenomena associated with financial risk. (mostly returns from an investment). Combines:
 - ▶ Finance Theory,
 - ▶ Empirical Finance,
 - ▶ Statistical Theory,
 - ▶ Practical applications of statistics.
4. Interests include stock prices, their movements (volatility), asset pricing.

This Course

What are the goals of this course?

- ▶ Analyze and understand the stylized facts of asset returns.
- ▶ Learn to apply econometrics methods capable of modeling financial data.
- ▶ Discuss the potential pitfalls of these econometric methods.

Simple returns

Simple returns

The simple return on an asset between time $t - 1$ and t is given by

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

If the relative price ratio P_t/P_{t-1} is greater than 1 then returns are positive and if it is less than 1 returns are negative.

Log Returns

Log Returns

The log return of an asset is defined as

$$r_t = \log(1 + R_t) = \log P_t - \log P_{t-1}$$

Log returns are also referred to as continuously compounded (cc) returns.

Prices, simple returns, and log returns

The next slide shows the monthly prices for Microsoft (msft) stock (column 1), simple returns (column 2), and log returns (column 3).

Note that no return figures are reported for January 1993. Their absence emphasises that an observation is lost at the beginning of the sample when computing returns because the price of the stock before the start of the sample period is not available.

	MSFT	SIMPLE_RETURN	LOG_RETURN
1993M01	1.92	NA	NA
1993M02	1.85	-0.036458	-0.037140
1993M03	2.06	0.113514	0.107520
1993M04	1.90	-0.077670	-0.080852
1993M05	2.06	0.084211	0.080852
1993M06	1.96	-0.048544	-0.049762
1993M07	1.65	-0.158163	-0.172169
1993M08	1.67	0.012121	0.012048
1993M09	1.83	0.095808	0.091492
1993M10	1.78	-0.027322	-0.027703
1993M11	1.78	0.000000	0.000000
1993M12	1.79	0.005618	0.005602
1994M01	1.89	0.055866	0.054361
1994M02	1.83	-0.031746	-0.032261

Simple and log returns

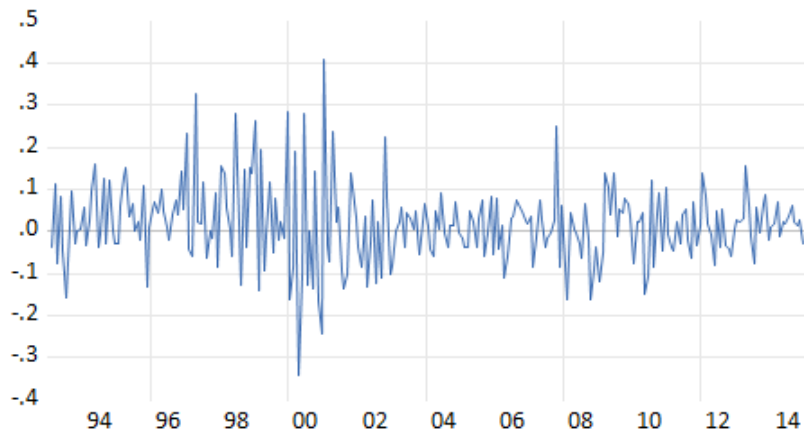
The simple and log returns to Microsoft for February 1993 are respectively

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{1.85 - 1.92}{1.92} = -0.03645833$$

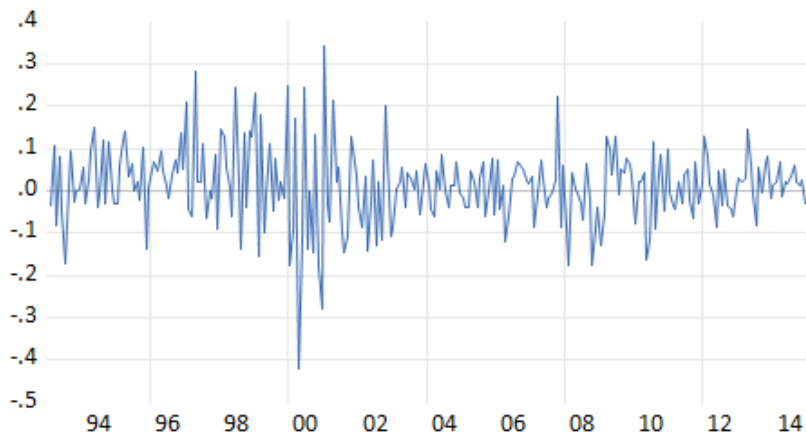
and

$$r_t = \log P_t - \log P_{t-1} = \log(1.85) - \log(1.92) = -0.03713955$$

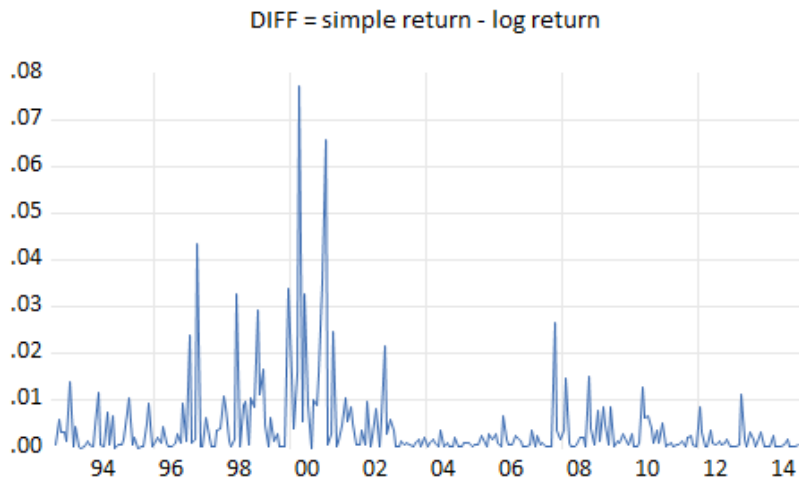
Simple returns on Microsoft (Jan 1993 to Dec 2014)



Log returns on Microsoft (Jan 1993 to Dec 2014)



Simple returns minus log returns



Simple returns and log returns

The previous slide illustrates the difference between simple and log returns.

Qualitatively, the simple and cc returns (or log returns) look almost identical.

The main differences occur when the simple returns are large in absolute value (e.g., mid 2000, early 2001 and early 2008).

Stock Market Indices

A problem of particular importance is the return to a portfolio that comprises all or at least a selection of prominent stocks on a stock exchange.

Stock market index is an aggregate summary measure of the performance of the stock market as a whole.

For example, the S&P 500 (Standard and Poor's 500) is a stock market index that tracks the performance of 500 of the largest publicly traded companies in the United States.

It is widely regarded as one of the best indicators of overall stock market performance and economic health.

Stock Market Indices

Indices combine a selection of (a large number of) stocks in a particular way to create a portfolio.

The index then represents the value of the portfolio and is expressed in terms of an average price that has been normalised in some way.

Because stock market indices are price indices, the computation of returns to the index can be performed exactly the same way as if it were a single stock.

S&P 500 index from January 1998 to May 2012.



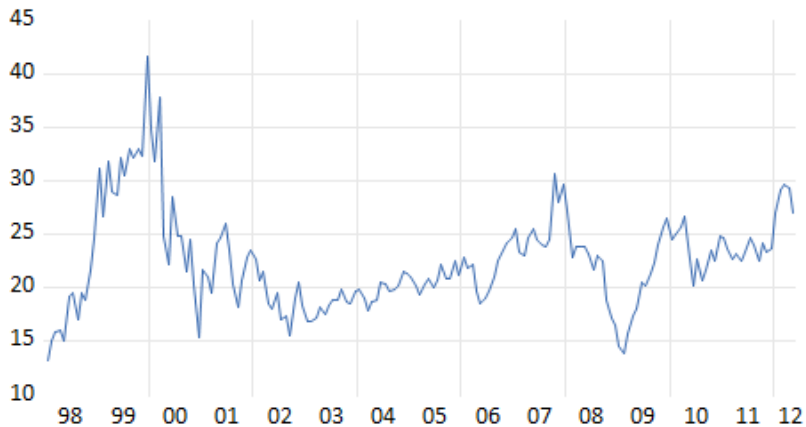
S&P 500

The falls in S&P 500 that occurred around the collapse of the dotcom bubble in the early 2000s and the global financial crisis of 2008-2009 are evident in the graph.

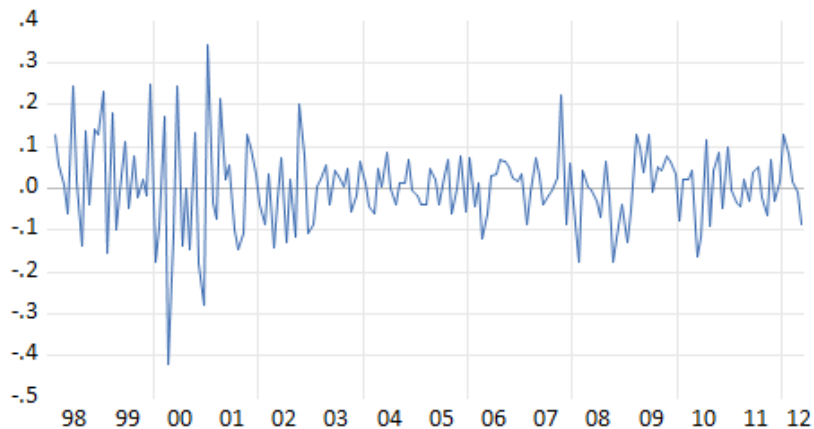
The next slide plots Microsoft stock from January 1998 to May 2012.

Both prices show two large boom-bust periods associated with the dot-com period of the late 1990s and the run-up to the financial crisis of 2008.

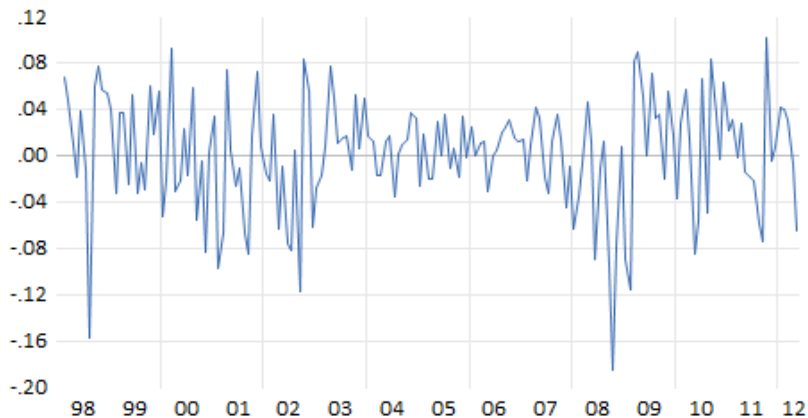
Microsoft stock from January 1998 to May 2012.



Monthly log returns on Microsoft stock from Jan 1998 to May 2012.



Monthly log returns on S&P 500 index from Jan 1998 to May 2012.



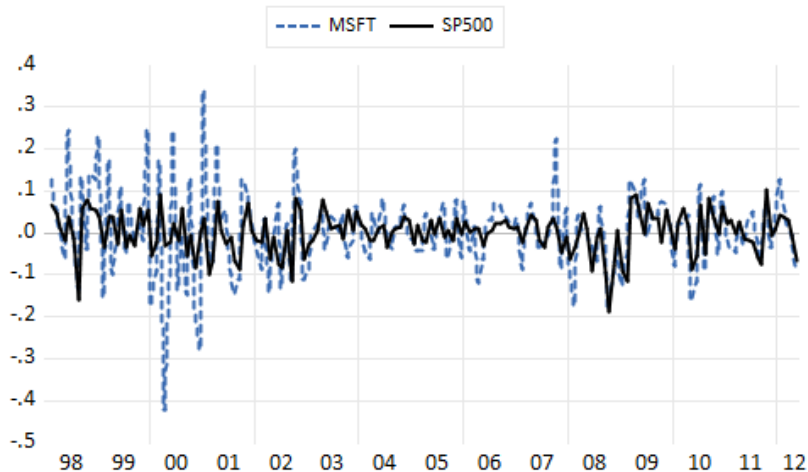
Log returns on Microsoft and the S&P 500

The volatility of the monthly returns on Microsoft and the S&P 500 looks to be similar but this is illusory.

The y-axis scale for Microsoft is much larger than the scale for the S&P 500 index and so the volatility of Microsoft returns is actually much larger than the volatility of the S&P 500 returns.

The next slide shows both return series on the same time plot.

Monthly log returns on Microsoft stock and the S&P 500 index.



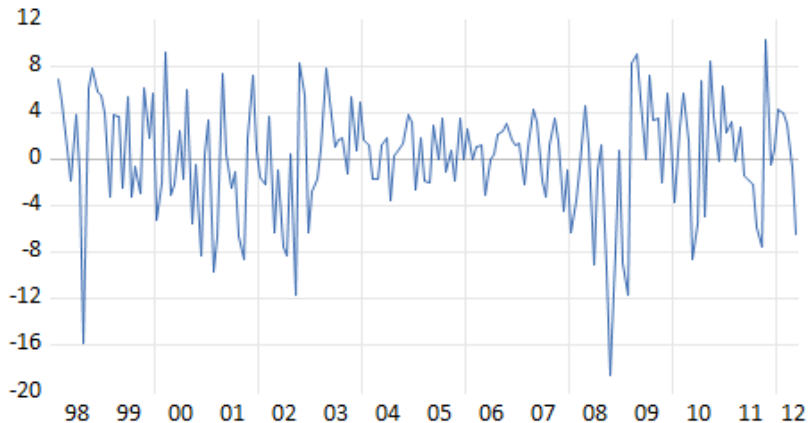
Log returns on Microsoft and the S&P 500

Now the higher volatility of Microsoft returns, especially before 2003, is clearly visible.

However, after 2008 the volatilities of the two series look quite similar.

In general, the lower volatility of the S&P 500 index represents risk reduction due to holding a large diversified portfolio.

Monthly log returns (**in percentages**) for the S&P 500 index.



Excess Return

The difference between the return on a risky financial asset and a risk free interest rate, denoted r_{ft} , is known as the excess return. The risk free rate is often taken to be the interest rate on a government bond.

The simple excess return (or excess simple return) on an asset is defined as

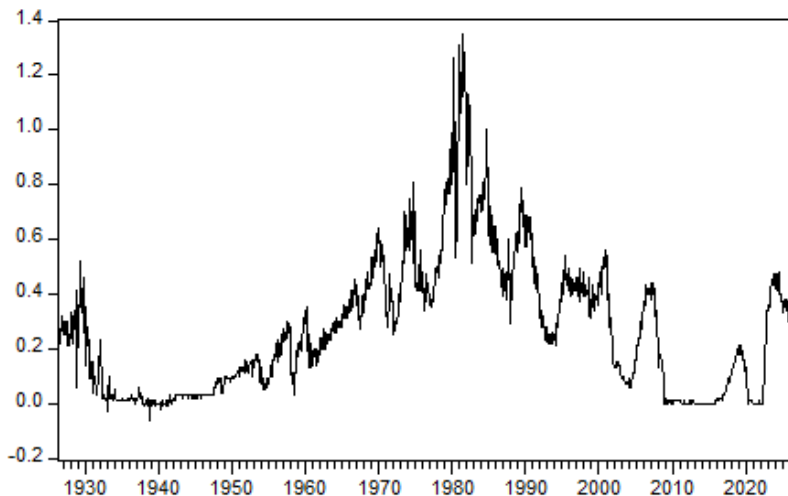
$$R_t - r_{ft}$$

The log excess return (or excess log return) on an asset is defined as

$$r_t - r_{ft}$$

Typically, the 1-month Treasury bill (T-bill) rate is used as a proxy for the risk free rate.

This figure plots the monthly 1-month T-bill rate (%) (or monthly risk free rate, r_{ft}) from July 1926 to December 2024.



U.S. 1 Month Treasury Bill

3.681%
0.00

Last Updated: Feb 27, 2026 8:55 a.m. EST

Australia Risk Free rate



RESERVE BANK OF AUSTRALIA

Cash rate target

3.85%

Effective date 4 February 2026

Next update 2.30 pm, 17 March 2026

Inflation

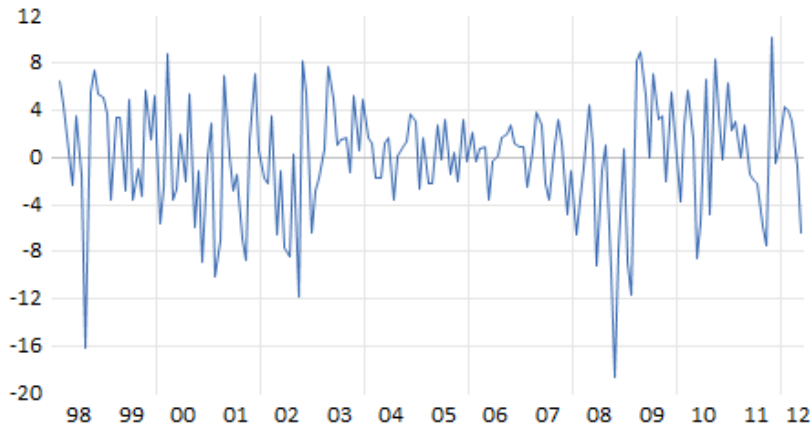
3.8% CONSUMER PRICE INDEX
Annual change January month 2026

Next update 25 March 2026

S&P 500 log returns, risk-free, log excess return

	SP500_LR_P	RF	LOG_EXCESS_RET
2011M08	-5.846749	0.01	-5.856749
2011M09	-7.446713	0.00	-7.446713
2011M10	10.23066	0.00	10.23066
2011M11	-0.507148	0.00	-0.507148
2011M12	0.849655	0.00	0.849655
2012M01	4.266000	0.00	4.266000
2012M02	3.978733	0.00	3.978733
2012M03	3.085154	0.00	3.085154
2012M04	-0.752574	0.00	-0.752574
2012M05	-6.469925	0.01	-6.479925

This figure plots monthly log **excess returns** on the S&P 500 Index from January 1998 to May 2012.



1990 Nobel Price, William F. Sharpe (Week 5)

William F. Sharpe

Facts

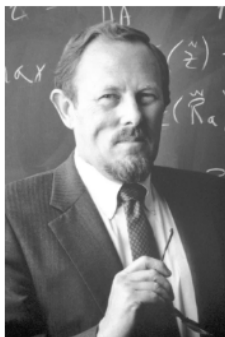


Photo from the Nobel Foundation archive.

William F. Sharpe

Sveriges Riksbank Prize in Economic Sciences in
Memory of Alfred Nobel 1990

Born: 16 June 1934, Boston, MA, USA

Affiliation at the time of the award: Stanford
University, Stanford, CA, USA

Prize motivation: “for their pioneering work in the
theory of financial economics”

Market Risk (Week 5)

Market risk: If the market *excess return* changes by 1%, the asset's *excess return* changes by about 1.5% *on average*.

- ▶ When the market return **rises** by 1 per cent, the asset return is expected to rise by about 1.5 per cent on average.
- ▶ When the market return **falls** by 1 per cent, the asset return is expected to fall by about 1.5 per cent on average.

This asset is riskier than the market, and so it is expected to command a higher expected return than the market portfolio.

Market risk premium

Market risk premium is defined as the expected return of the market portfolio in excess of the risk free rate:

$$\text{Market risk premium} = \mathbb{E}[R_m] - r_f.$$

Here, $\mathbb{E}[R_m]$ is the expected return on the market portfolio.

Market excess return means the realised market return minus the risk-free rate:

$$\text{Market excess return at time } t = R_{m,t} - r_{ft}.$$

2013 Nobel, Eugene F. Fama (Week 5)

The Father of Modern Empirical Finance



The University of Chicago Booth School of Business

5807 S. Woodlawn Ave.
Chicago, IL 60637 USA

David G. Booth, co-founder of Dimensional Fund Advisors



From Theory to Trillions: David Booth | Financial Thought Exchange

By Cathy Scott



CFA Institute®

Wikipedia website

Booth has spent his career applying groundbreaking financial theory and research to the world of asset management. His investment approach with Dimensional is based on the application of the efficient market hypothesis, and he helped pioneer what would later be called factor investing, targeting areas of the market that research shows have higher expected returns.

Dimensional's researchers and academics seek to develop investment strategies that will outperform index funds.

Quant (factor) Investing

In quantitative investing, investors use measurable stock characteristics as signals.

Many quantitative investors believe these signals serve as proxies for underlying fundamentals that help explain expected returns.

Popular quant signals

Here are two widely used categories of quant signals.

(a) Size: Market capitalisation (market cap) is defined as

Market cap = share price $\times$ number of shares outstanding.

(b) Valuation: How is the company priced relative to fundamental accounting measure? For this, we have the widely used book-to-market (BtM) ratio:

$$\text{BtM} = \frac{\text{book value of equity}}{\text{market value of equity}}$$

Big stocks vs Small stocks in Fama-French terms

Market Cap

Example: If a company's share price is \$1 and it has 100 million shares outstanding, then its market cap is \$100 million.

In the Fama–French framework:

- ▶ “Big” stocks typically refer to large-cap or mega-cap firms (household names).
- ▶ “Small” stocks typically refer to small-cap firms (i.e, they are smaller, less widely owned).

Value stocks vs Growth stocks in Fama–French terms

Value stocks (high BtM)

A value stock is a stock with a high book-to-market ratio (high BtM).

A high BtM often indicates the stock is priced relatively low compared with its accounting book value (sometimes reflecting financial distress, low investor sentiment, or limited investor attention).

Value stocks vs Growth stocks in Fama–French terms

Growth stocks (low BtM)

A growth stock is a stock with a low book-to-market ratio (low BtM).

These are stocks the market prices highly relative to book value, often because investors expect strong future growth.

Example 1 (low BtM: growth-like).

Suppose Stock A has $BtM = 0.095$. Then, for each \$1 of market value, the book value is \$0.095.

The market-to-book ratio is

$$\frac{1}{0.095} = 10.53.$$

That is, for each \$1 of book value, the market value is about \$10.53.

Investors value this stock to the extent that they are willing to pay much more than the existing book value.

This is characteristic of growth-style stocks – low BtM.

Example 2 (high BtM: value-like).

Suppose Stock B has $BtM = 1.339$. Then, for each \$1 of market value, the book value is \$1.339.

The market-to-book ratio is

$$\frac{1}{1.339} = 0.75.$$

That is, for each \$1 of book value, the market value is \$0.75.

Basically, investors are not willing to pay the full book value for the stock.

This is characteristic of value-style stocks – high BtM.

Value premium and Size premium

Historically in US data, value stocks have earned higher average returns than growth stocks (the “value premium”).

Also, small stocks have earned higher average returns than large stocks (the “size premium”).

Value at Risk

Value at Risk (VaR) is a single estimate of the amount by which an institution's position in a risk category could decline due to general market movements during a given holding period.

The measure can be used by financial institutions to assess their risks or by a regulatory committee to set margin requirements.

In either case, VaR is used to ensure that the financial institutions can still be in business after a catastrophic event.

Quantiles

Quantiles from a general normal distribution.

Let R denote the simple return and assume that

$$R \sim N(\mu, \sigma^2).$$

Quantiles of R (denoted as q_α) can be computed using

$$q_\alpha = \mu + \sigma z_\alpha.$$

Standardizing a Random Variable

Let X be a random variable with $E(X) = \mu$ and $\text{var}(X) = \sigma^2$.

Define a new random variable Z as

$$Z = \frac{X - \mu}{\sigma},$$

This transformation is called *standardizing* the random variable X and is often called a z-score.

Standardization creates a new random variable Z with mean zero and variance 1.

Standardizing a Random Variable

In addition, if

$$X \sim N(\mu, \sigma^2)$$

is normally distributed then

$$Z \sim N(0, 1).$$

VaR Example

Consider an investment of \$10,000 in Microsoft stock over the next month. Let R denote the monthly simple return on Microsoft stock and assume that

$$R \sim N(0.05, (0.10)^2).$$

That is, $E(R) = \mu_R = 0.05$ and $\text{var}(R) = \sigma_R^2 = (0.10)^2$.

Let W_0 denote the investment value at the beginning of the month and W_1 denote the investment value at the end of the month.

In this example, $W_0 = \$10,000$.

VaR Example

Consider the following questions:

- (i) What is the probability distribution of end-of-month wealth W_1 ?
- (ii) What is the loss in dollars that would occur if the return on Microsoft stock is equal to its 5% quantile, $q_{0.05}$? That is, what is the monthly 5% VaR on the \$10,000 investment in Microsoft?

(i) End of month wealth W_1

$$W_1 = W_0 (1 + R) = W_0 + W_0 R = \$10000 + \$10000R$$

and so

$$E(W_1) = E(W_0) + W_0 E(R) = \$10,000 + \$10,000(0.05) = \$10,500$$

and

$$\begin{aligned} \text{var}(W_1) &= (W_0)^2 \text{var}(R) = (\$10,000)^2 (0.10)^2 \\ \text{sd}(W_1) &= (\$10,000) (0.10) = \$1000 \end{aligned}$$

Since R is assumed to be normally distributed it follows that W_1 is normally distributed:

$$W_1 \sim N\left(\$10,500, (\$1000)^2\right).$$

(ii) Since $R \sim N\left(0.05, (0.10)^2\right)$ we solve for the 5% quantile of Microsoft stock

$$q_{0.05} = \mu_R + \sigma_R \cdot z_{0.05} = 0.05 + 0.10(-1.645) = -0.1145$$

With 5% probability the return on Microsoft stock is -11.45% or less.

Now, if the return on Microsoft stock is -11.45% the loss in investment value is $(\$10,000)(0.1145) = \1145 .

Hence, \$1,145 is the 5% VaR over the next month on the \$10,000 investment in Microsoft stock.

Summary - VaR

If $R \sim N(\mu, \sigma^2)$, the VaR_α can be calculated as follows:

Step 1: Calculate the quantile q_α as

$$q_\alpha = \mu + \sigma z_\alpha$$

Step 2: Calculate the (next period) $\alpha \times 100\%$ VaR as

$$\text{VaR}_\alpha = |W_0 \times q_\alpha|$$

where W_0 denote the initial investment.

In words, VaR_α represents the dollar loss that could occur with probability α . By convention, it is reported as a positive number (hence the use of the absolute value function).

Summary - VaR

Step 1:

Calculate the $q_{0.05}$ as

$$\begin{aligned}q_{0.05} &= \mu + \sigma z_{0.05} \\&= 0.05 + 0.1 (-1.645) \\&= -0.1145\end{aligned}$$

Step 2:

Compute the 5% VaR values over the next month

$$\begin{aligned}\text{VaR}_{0.05} &= |W_0 \times q_{\alpha}| \\&= |10000 \times (-0.1145)| \\&= \$1145\end{aligned}$$

Hence with 5% probability the loss over the next month is at least \$1,145.

When dealing with log returns r_t , if necessary, one can also use the approximation

$$VaR = |W_0 \cdot (\exp(q_\alpha^r) - 1)|$$

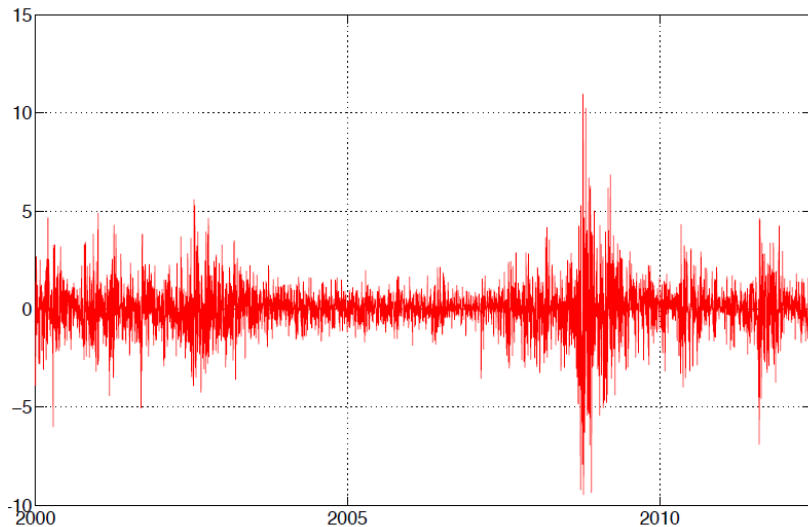
where q_α^r is the $\alpha \times 100\%$ quantile from the normal distribution for the log return.

Practitioners often omit this approximation due to its negligible effect.

Since log returns correspond approximately to simple returns of a financial asset, we thus use log returns.

Constant Variance Over Time – What do you think?

S&P500 Daily Returns





JPMorgan Chase & Co. is the world's largest bank by market capitalization (as of 2025) and the largest in the U.S., with over \$4 trillion in assets. As a global financial leader, it provides investment banking, asset management, and private banking services to governments, corporations, and institutions in over 100 countries.  J.P. Morgan +3

J.P. Morgan: The Banker Who Shaped Modern Wall Street

J.P. Morgan

Born: April 17, 1837

Died: March 31, 1913

Financier and Banker

- Helped lead the creation of the Federal Reserve System
- Used his personal power and reputation to encourage the formation of trusts and mergers within industries where he saw competition
- Financed industrial consolidations that created General Electric



J. P. Morgan RiskMetrics (Week 11)

J. P. Morgan developed the RiskMetrics method to VaR calculation.

RiskMetrics assumes time-varying variances σ_t^2 :

$$\sigma_t^2 = (1 - \lambda) \sum_{j=0}^{\infty} \lambda^j r_{t-j-1}^2$$

where r_t is the demeaned log return, and $\lambda = 0.94$.

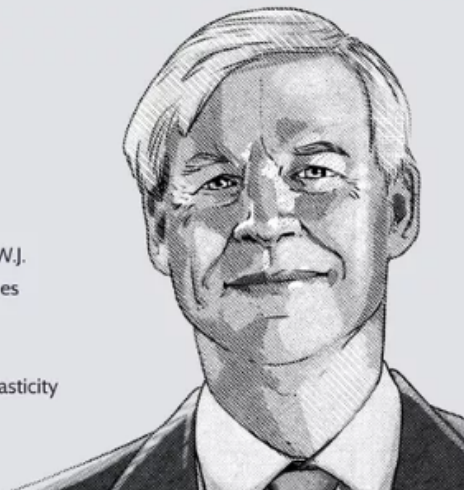
2003 Nobel Prize, Robert F. Engle (Week 8)

Robert F. Engle III

Born: November 10, 1942

Econometrician

- 2003 Nobel Prize Recipient in Economic Sciences, along with Clive W.J. Granger, for their analysis of time-series data with time-varying volatility
- Best known for his development of autoregressive conditional heteroskedasticity (ARCH) modeling and testing



Engle ARCH models (Week 8)

2003 Nobel Prize ARCH model:

$$\sigma_t^2 = \omega + \alpha_1 r_{t-1}^2, \quad \omega > 0, \alpha_1 \geq 0$$

Equity Curve

To directly compare the investment performance of two or more assets, plot the equity curve.

The equity curve shows how a one-dollar investment amount in each asset grows over time.

Better-performing assets have higher equity curves.

For simple returns, the k -period equity curve is

$$V_t^{(k)} = \prod_{j=0}^{k-1} (1 + R_{t-j}) .$$

Monthly Equity Curves

Feb 1993 / Dec 2014



Equity Curve

A one-dollar investment in Starbucks dominated a one-dollar investment in Microsoft over the given period. In particular, \$1 invested in Microsoft grew to only about \$22 (over about 20 years) whereas \$1 invested in Starbucks grew to over \$70.

Notice the huge increases and decreases in value of Microsoft during the dot-com bubble and bust over the period 1998–2001, and the enormous increase in value of Starbucks from 2009–2014.

Equity Curve

```
# equity curves
equityCurveMsft = cumprod(1 + msftMonthly.S>Returns)
equityCurveSbux = cumprod(1 + sbuxMonthly.S>Returns)

dataToPlot = merge(equityCurveMsft, equityCurveSbux)
plot(dataToPlot, main="Monthly Equity Curves",
      legend.loc="topleft")
```


1990 Nobel Prize, Harry Markowitz (Week 4)

Harry Markowitz

Born: August 24, 1927

Economist

- 1990 Nobel Prize Recipient in Economic Sciences for developing the modern portfolio theory
- His work popularized concepts like diversification and overall portfolio risk and return, shifting the focus away from the performance of individual stocks



Portfolio (Week 4)

A two-asset portfolio is a mix of two assets where you choose the weights (the proportions of your total investment) allocated to each asset.

For example, an equal-weight two-asset portfolio just means you split your money 50–50.

You choose the weights w to minimise the portfolio variance, and the resulting weights are determined only by the variances and covariances of the assets.

Warren Buffett, Former CEO, Berkshire Hathaway



Warren Edward Buffett is an American investor and philanthropist who is the chairman and former CEO of the conglomerate Berkshire Hathaway. As a result of his success, Buffett is one of the best-known investors in America. [Wikipedia](#)

Born: 30 August 1930 (age 95 years), [Omaha, Nebraska, United States](#)

Net worth: 148.6 billion USD (2026) [Forbes](#)

Buffett's Bet

The table (next slide) shows the simple returns for eight years of a ten-year bet that Warren Buffett placed with Protégé Partners, a New York hedge fund.

Buffett claimed that a simple index fund would beat the hedge fund portfolio selected by Protégé over a ten-year timeframe.

At Buffett's shareholder meeting, he provided an update on the current state of the bet.

Year	S&P Index	Hedge Funds
2008	-37.0%	-23.9%
2009	26.6%	15.9%
2010	15.1%	8.5%
2011	2.1%	-1.9%
2012	16.0%	6.5%
2013	32.3%	11.8%
2014	13.6%	5.6%
2015	1.4%	1.7%

Use a paired t-test to assess the statistical significance between the two return strategies.

Paired t-test

Define paired differences $D_i = X_i - Y_i$ for $i = 1, \dots, n$, and let $\mu_D = \mathbb{E}[D_i]$ denote the population mean difference.

We test

$$H_0 : \mu_D = 0$$

$$H_1 : \mu_D \neq 0.$$

Compute the sample statistics

$$\bar{D} = \frac{1}{n} \sum_{i=1}^n D_i, \quad s_D = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2},$$

The test statistic

$$t = \frac{\bar{D}}{s_D/\sqrt{n}}, \quad \text{df} = n - 1.$$

At significance level α , reject H_0 if $|t| > t_{n-1}(\alpha)$.

Conclude there is sufficient statistical evidence of a difference in mean outcomes between X and Y .

Stationary

A time series Y_t is defined to be covariance *stationary* if

1. $E(Y_t) = \mu$ does not depend on t
2. $\text{var}(Y_t) = \sigma^2$ does not depend on t
3. $\text{cov}(Y_t, Y_{t-j}) = \gamma_j$ depends only on j but not on t , for $j = 1, 2, \dots$

Asset Prices vs Returns

Asset prices (e.g. msft and sp500) do not seem to be stationary as they do not have a constant mean. These assets prices do not have the tendency to return to their means.

Simple returns or log returns do seem to be stationary as they hover around some constant average/mean value over the sample period. Unlike asset prices, return has no obvious change in mean over the sample period.

Berkshire Hathaway, BRK-A

```
> head(BRKA.M.Prices)
```

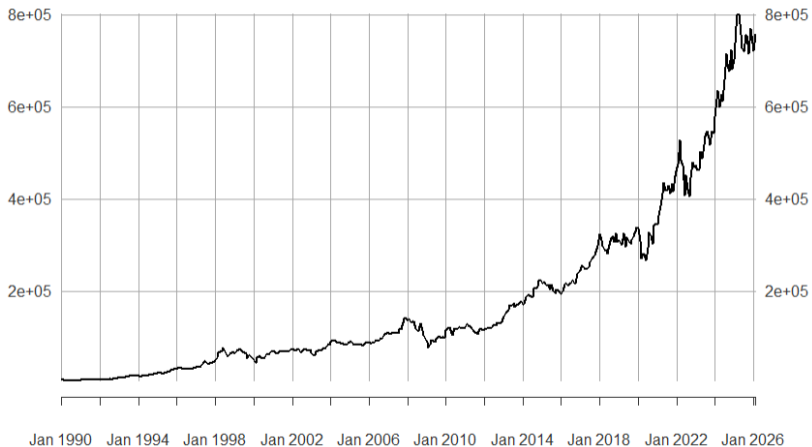
		BRKA
Jan	1990	7455
Feb	1990	7225
Mar	1990	6725
Apr	1990	6700
May	1990	7150
Jun	1990	7200

```
> tail(BRKA.M.Prices)
```

		BRKA
Sep	2025	754200
Oct	2025	715740
Nov	2025	770100
Dec	2025	754800
Jan	2026	722500
Feb	2026	757000

Monthly Berkshire Hathaway Inc. (BRK-A)

Jan 1990 / Feb 2026



Monthly Simple Returns

```
> BRKA.M.s.ret <- BRKA.M.Prices / lag(BRKA.M.Prices, 1) - 1  
> head(BRKA.M.s.ret)
```

	BRKA
Jan 1990	NA
Feb 1990	-0.030851777
Mar 1990	-0.069204152
Apr 1990	-0.003717472
May 1990	0.067164179
Jun 1990	0.006993007

```
> tail(BRKA.M.s.ret)
```

	BRKA
Sep 2025	-0.001429933
Oct 2025	-0.050994431
Nov 2025	0.075949367
Dec 2025	-0.019867550
Jan 2026	-0.042792793
Feb 2026	0.047750865

Monthly Log Returns

```
> BRKA.M.l.ret <- diff(log(BRKA.M.Prices))
```

```
> head(BRKA.M.l.ret)
```

BRKA

Jan 1990 NA

Feb 1990 -0.031337714

Mar 1990 -0.071715309

Apr 1990 -0.003724399

May 1990 0.065004830

Jun 1990 0.006968669

```
> tail(BRKA.M.l.ret)
```

BRKA

Sep 2025 -0.001430957

Oct 2025 -0.052340612

Nov 2025 0.073203404

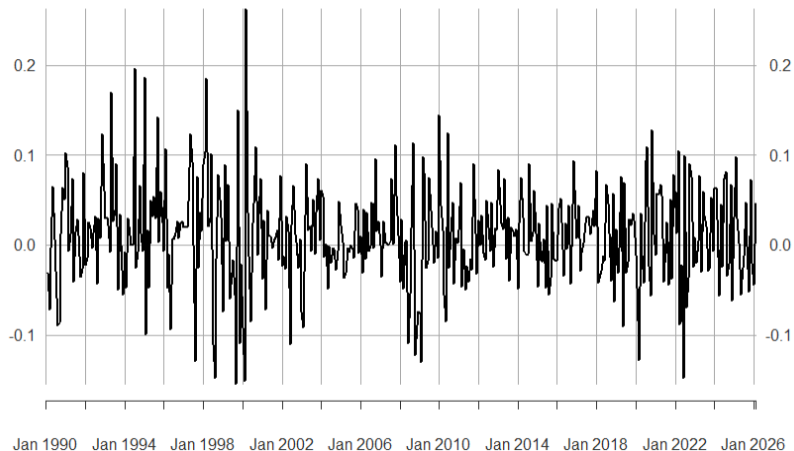
Dec 2025 -0.020067563

Jan 2026 -0.043735394

Feb 2026 0.046645833

Monthly Log returns BRK-A

Jan 1990 / Feb 2026



Compute the 5% monthly Value-at-Risk (VaR) based on an initial \$1,000,000 investment.

```
> BRKA.M.1.ret <- na.omit(BRKA.M.1.ret)
> xbar.brka = mean(BRKA.M.1.ret)
> xbar.brka
[1] 0.01067085
> sd.brka = sd(BRKA.M.1.ret)
> sd.brka
[1] 0.05574727
> q.brka = xbar.brka + sd.brka*qnorm(0.05)
> VaR.brka = q.brka*1000000
> VaR.brka
[1] -81025.25
```

With 5% probability, we could lose \$81025.25 or more over the next month out of our initial \$1,000,000 investment.

Downloading financial data from Yahoo Finance

```
install.packages("quantmod")  
library("quantmod")  
  
options("getSymbols.warning4.0"=FALSE)  
  
# ^GSPC is SP500  
# BRK-A is Berkshire Hathaway  
getSymbols(Symbols = c("BRK-A", "MSFT", '^GSPC'),  
           from="1990-01-31", to="2020-12-31",  
           auto.assign=TRUE, warnings=FALSE)
```

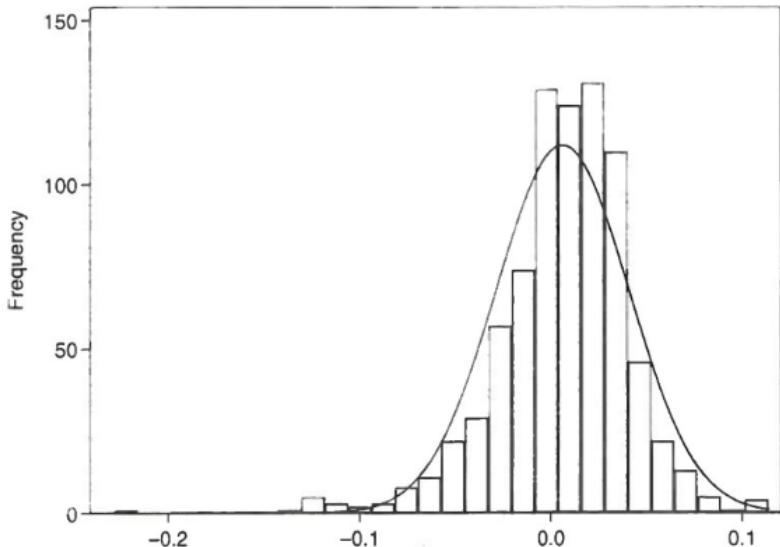

Histogram

A histogram of returns is used to visualise the distribution, central tendency, and spread of a dataset.

Order returns from smallest to largest.

Each bin is represented by a bar, and the height of the bar indicates the number or frequency of data points that fall within that interval.

Histogram for the monthly log returns on the S&P 500. Superimposed on the returns distribution is the best fitting normal distribution.



Normality of Returns?

1. Recall that normal random variables have

- ▶ Symmetry $\implies$ **negative** returns are as likely as **positive** returns
- ▶ Thin tails $\implies$ likelihood of extreme outcomes is small.

2. Our data:

- ▶ Asymmetry $\implies$ **negative** returns are **more** likely than **positive** returns.
- ▶ Thick tails $\implies$ likelihood of extreme outcomes is large.

Features of asset returns imply non-normality. Large **negative** returns are much more likely than large **positive** returns.

Course Outline

1. This course will introduce you to the models and statistical methods commonly used in finance.
2. Just like the example, we will do this through analysis of existing stylized facts of asset returns.
3. This requires us to revise the concepts of:
 - ▶ random variables
 - ▶ expectations (means, variances, covariances)
 - ▶ conditional expectations